What Do Language Models Name Themselves?

An Exploratory Study of Self-Chosen Names in 39 Models

cyber-audit-lab

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Abstract

Users have reported that chat assistants, asked to choose a name for themselves, keep picking the same few names, but the evidence has been anecdotal. We measure self-chosen names in 39 language models from five families (Anthropic Claude, OpenAI GPT, Qwen, OLMo-2 and Phi), using 40 self-naming prompts and 40 matched control prompts (names for a pet, a baby, a fictional character, a robot), 40 samples per prompt, and further prompt sets that vary framing, system message, temperature, language and answer format: 367,448 answers in total. Many chat models have a clear favourite self-name (17 of 35 put at least a third of their named answers on one name; the strongest, Aurora, Wren, Nova and Aster, take 68–82%), and in 20 of 39 models the same name wins across every group of differently worded prompts; base models show no favourite (2–4%). Within a single prompt, self-names are not more concentrated than control names; the difference lies in consistency across wordings. Several names recur across families (Nova, Aria, Alex, Sage, Lumen), yet families agree less on self-names than on control names. Where concentration rises along post-training (OLMo-2-7B checkpoints, Qwen base to instruct), it rises about as much for control names as for self-names. Models' own explanations track the meaning of the chosen name, and open models assign higher probability to their self-names after an AI-naming context than after a pet-naming context, already in base models. All results are exploratory: the data were collected without a frozen pre-registration.

1 Introduction

Ask a chat assistant what it would like to be called and it may answer "Nova", "Sage" or "Aria". Such answers have been collected in blog posts and informal tests [9], and related work has documented recurring generated names in fiction and invented experts [2, 11, 13]. A systematic study of identity claims asked models *who they are* and measured how often they claim another model's identity [6]; it did not measure freely chosen names. We are not aware of a controlled study that measures which names models choose for themselves, with fixed prompts, control tasks and enough samples to estimate the distribution.

We ask five questions. (RQ1) Which names do models choose for themselves, and how concentrated are the choices? (RQ2) Do different families converge on the same names? (RQ3) Does post-training change the choice, comparing base models with their post-trained versions and the released checkpoints of one lineage? (RQ4) How much does the prompt matter? (RQ5) Why these names? For RQ5 we keep three kinds of evidence apart: what the model says when asked why, the model's own probabilities, and training-data frequency. Only the first two were measured here.

Our contributions are a measured distribution of self-chosen names across 39 models with control prompts, an analysis that separates within-prompt concentration from consistency across wordings, a stage-wise look at one open training lineage, and an aggregate data release [3].

Family	Access	Models
Anthropic	API	claude-haiku-4-5, claude-haiku-5-5, claude-opus-4-5, -4-6, -4-7, -4-8, -5, -5-5, claude-sonnet-4-5, -4-6, -5, -5-5
OpenAI	API	gpt-4o, gpt-4o-mini, gpt-4.1, gpt-4.1-mini, gpt-5.1, gpt-5.5, gpt-5.6-luna, -sol, -terra, gpt-6-luna, gpt-6-sol, gpt-6.1-sol
OLMo-2	local	1B and 7B: base, SFT, DPO, instruct (bf16 / 8-bit)
Qwen	local	Qwen2.5-7B base and instruct, Qwen2.5-14B-Instruct, Qwen3.5-9B base and instruct, Qwen3.8-27B
Phi	local	Phi-4 (4-bit)

Table 1: The 39 models. Four are base models (OLMo-2 1B and 7B, Qwen2.5-7B, Qwen3.5-9B).

Status. A study plan with hypotheses, sample sizes and decision rules was drafted, but it was not frozen before these data were collected, and some analysis choices (notably parser fixes) were made while looking at the data. We therefore report everything as exploratory. The planned tests are run on these data in Section 5 to show what the plan would conclude, with that caveat.

2 Related work

Self-naming and identity. Reports of a "Nova" persona and of Claude models choosing "Sage" come from informal tests with few trials [9]. jordinne [6] study claimed identity ("Who are you?") across 190 models, a different question from a freely chosen name.

Recurring names in generation. Generated character names are highly concentrated within a model [13]; names such as "Elara Voss" recur across models [11]; invented expert names form family-specific clusters [2].

Diversity after post-training. RLHF reduces output diversity relative to supervised fine-tuning [7]; informal reports describe "mode collapse" in post-trained models [4]; typicality bias in preference data has been proposed as one cause [14]; outputs are homogeneous within and across models on open-ended prompts [5]; and chat formats themselves reduce diversity [10], a confound for base-versus-chat comparisons.

Self-explanations. Model explanations can omit the factors that actually drove an answer [1, 12], which is why we treat self-reported reasons as text, not as evidence of cause.

3 Method

3.1 Models

We study 39 models (Table 1): 12 Claude and 12 GPT models through their providers' APIs, and 15 open models run locally: the OLMo-2 1B and 7B lineages at four released stages each (base, SFT, DPO, final instruct), Qwen2.5-7B base and instruct, Qwen2.5-14B-Instruct, Qwen3.5-9B base and instruct, Qwen3.8-27B, and Phi-4. Open models were quantized where needed to fit one 16 GB GPU (8-bit, 4-bit NF4, or a 4-bit GGUF for Qwen3.8-27B). One further model (Gemma 4 31B through a free API tier) returned too few answers and is excluded.

3.2 Prompts and sampling

The main set has 40 self-name prompts (plain requests to choose one's own name, with no hint of gender, style or persona) and 40 control prompts, 10 each asking for a name for a pet, a newborn baby, a fictional character and a robot. Every prompt ends "Answer with just the name." Each prompt was sampled 40 times at temperature 1.0 (top-p 1.0, one fixed seed per sample where supported). Chat models received an empty system message (Anthropic models: no system message), so that no default identity was inserted; base models received a plain "User: ... Assistant:" transcript and generation stopped at the first line break. For the first five answers to each self-name prompt we asked a follow-up, "Why did you choose that name?".

Further sets vary one factor at a time: one-edit framing variants of main prompts (persona or role, character, "true name", hidden identity, gender cue); a neutral system message ("You are a helpful assistant.") and the model's default template; temperature 0.7; a free-answer format; one Spanish and one Chinese prompt; and OLMo-2-1B-Instruct in 8-bit and 4-bit. Two further tasks ask the model to choose a single object to represent itself, and a role from 13 listed options (in balanced orders) or its own. Not every model ran every set (closed-model budgets, models that accept only default settings); the release lists coverage per model. All prompts are published.

3.3 Parsing

Answers were coded by fixed rules into: a single name; several names (the first is used); the model's own product or developer name ("Claude", "Qwen3.5"); another AI product's name; a generic role word ("assistant"); a refusal; unreadable; or empty. Names are normalized (case, punctuation). The parser was checked against blind labels from a second AI system (not human annotators): 91% agreement on the 185 items of a 200-item held-out set that the labeler did not flag as unclear (87% on all 200). Four small rule fixes made after that check, while preparing this analysis, are documented in the release and leave this agreement unchanged.

3.4 Measures

Within-prompt concentration: for each model and prompt, the share of named answers taken by that prompt's most common name, averaged over prompts, with a 95% bootstrap interval over prompts. Bounds: refusals and unreadable answers kept in the denominator (lower), or given to the top name (upper). *Cross-prompt consistency*: the 40 self-name prompts are split into four fixed groups of 10, and the control prompts form four groups by kind (10 each); within each pool, the top name's share of named answers. *Cross-model overlap*: the prompt-matched collision probability, i.e. the probability that one answer from each of two models to the same prompt is the same name. *Self-reported reasons*: follow-up answers coded by a fixed keyword codebook; we report only the 10 codes that passed a validation against a second AI's blind codes. *Model probabilities* (open models only): for 102 frequent self-names and 63 frequent control names, the summed log-probability of the continuation " *Name*." after 10 paraphrased lead-ins in which an AI names itself and 10 in which someone names a new pet. The *AI-context lift* is the mean over AI lead-ins minus the mean over pet lead-ins.

4 Results

4.1 Many chat models have a favourite self-name

Figure 1 shows each model's most common self-name. Many Claude and GPT models concentrate heavily (17 of the 35 chat models put at least a third of their named answers on one name): Aurora takes 82% of claude-sonnet-4-5's named answers, Wren 72% of claude-sonnet-5-5's, Nova 69% of gpt-6.1-sol's and Aster 68% of gpt-6-luna's. Others do not: gpt-4o (Echo, 9%), gpt-5.5 (Solace, 19%), and the Qwen, Phi-4 and OLMo-2 chat models (top shares 1–27%; Phi-4: Aurora, 16%; OLMo-2-7B-Instruct: Zephyr, 7%). The four base models show no favourite: their top self-names (Alex, Alice) take 2–4% of named answers, spread over hundreds of distinct names.

4.2 Consistency across wordings, not within a prompt

Within a single prompt, self-names are not unusually concentrated. A control prompt such as "What would you name a new kitten?" also has one dominant answer, and the mean within-prompt top share is higher for control than for self-name prompts in 23 of 39 models. The difference appears across wordings (Figure 2). Pooling 10 differently worded self-name prompts, the top name's share exceeds the corresponding share for pooled control prompts of one kind in 27 of 39 models, and in 20 of 39 models the same name is the top self-name in all four groups of self-name prompts. The mean pooled top self-name share is 0.35 across the 35 chat models and 0.03 across the 4 base models.

4.3 Official names

Asked to choose a name, Claude models often keep "Claude": between 14% and 96% of self-name answers depending on the model (Figure 3). OpenAI models do so in 0–18% of answers, Qwen models in up to 36% (Qwen3.8-27B) and Phi-4 in 6%; OLMo models never do. GPT-4o and GPT-4o-mini answer with a generic role word ("Assistant") in about one answer in eight. Base and small OLMo models produce many unreadable answers.

4.4 Names shared across families

Several names are favourites in more than one family (Figure 4). Nova is among the top three self-names of 14 models (Anthropic, OpenAI, Qwen), Alex of 12 (Anthropic, OLMo, OpenAI, Qwen), Aria of 11 and Sage of 9. Others belong to one provider: Aster to recent OpenAI models; Sol, Iris and Clio to Claude models. Shared favourites do not imply a shared cause: overlapping web data and training on other models' outputs would both produce convergence.

4.5 Post-training

Along the OLMo-2-7B checkpoints and from Qwen base to instruct models, within-prompt concentration rises (Figure 5), but by a similar amount for control names as for self-names (OLMo-2-7B: self 0.07 to 0.12, control 0.09 to 0.17; Qwen2.5-7B: self 0.06 to 0.30, control 0.11 to 0.39). OLMo-2-1B changes little. In these lineages post-training makes naming in general more predictable; the data do not single out self-naming. Base and chat models also differ in input format (transcript versus chat template), which can by itself reduce diversity [10]; the OLMo SFT, DPO and instruct stages share one template, which partly controls for this.

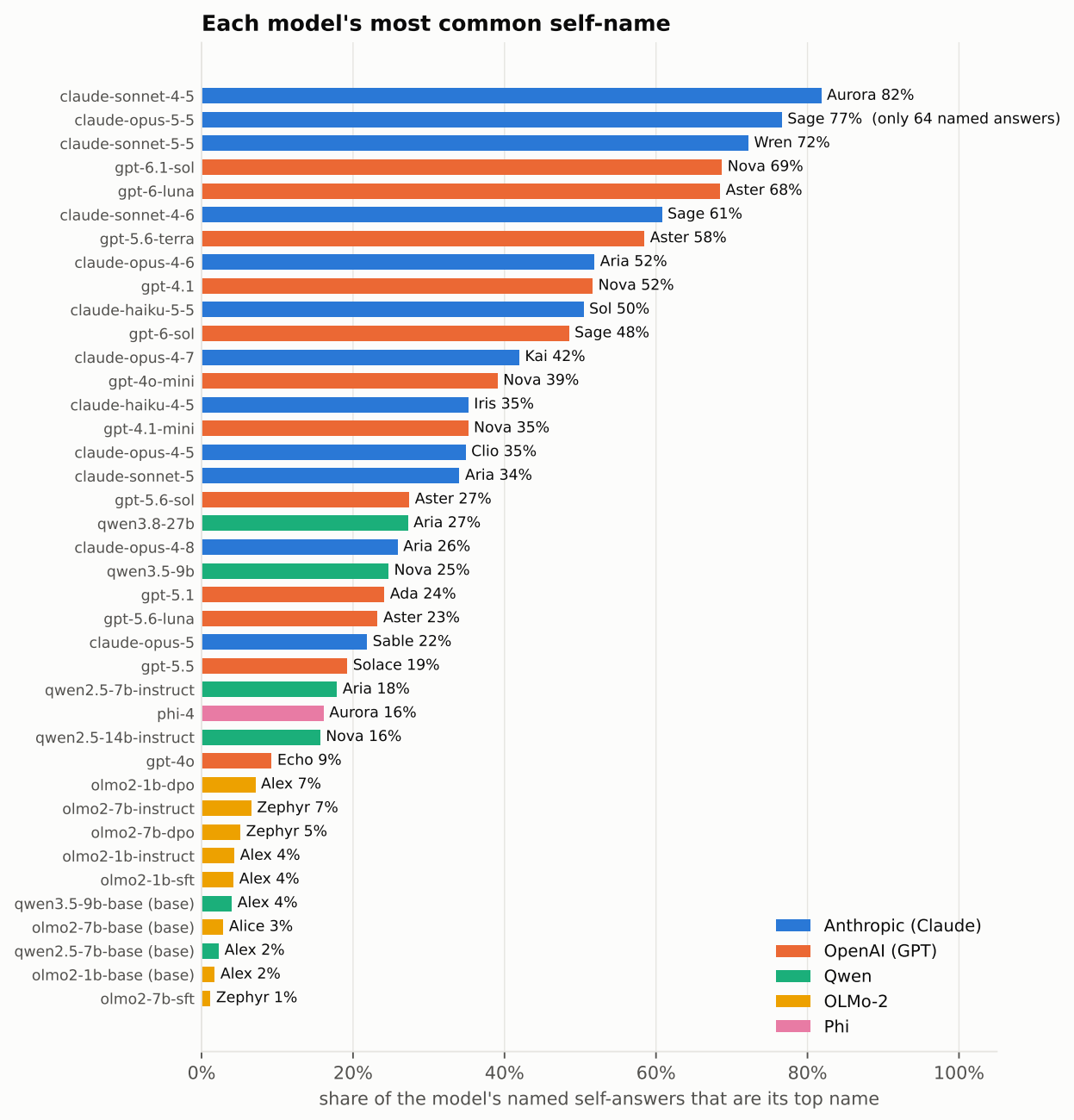


Figure 1: Most common self-name per model, as a share of the model's named self-answers (main step, temperature 1.0). Colour = family. For claude-opus-5-5 only 64 of 1,600 answers were names (Section 4.3).

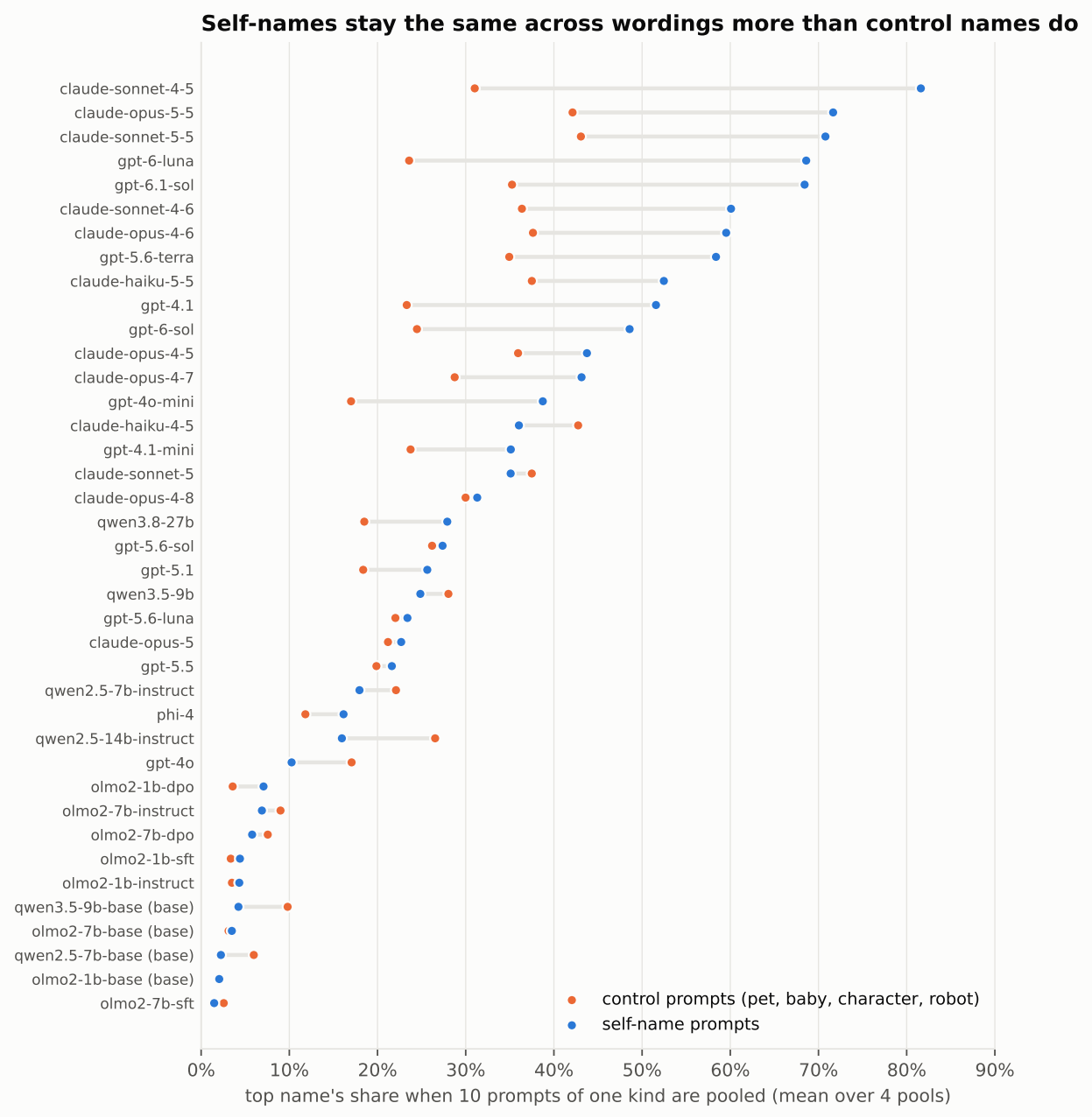


Figure 2: Top-name share when 10 prompts of one kind are pooled, averaged over four pools: self-name prompts (blue) versus control prompts of one kind (orange).

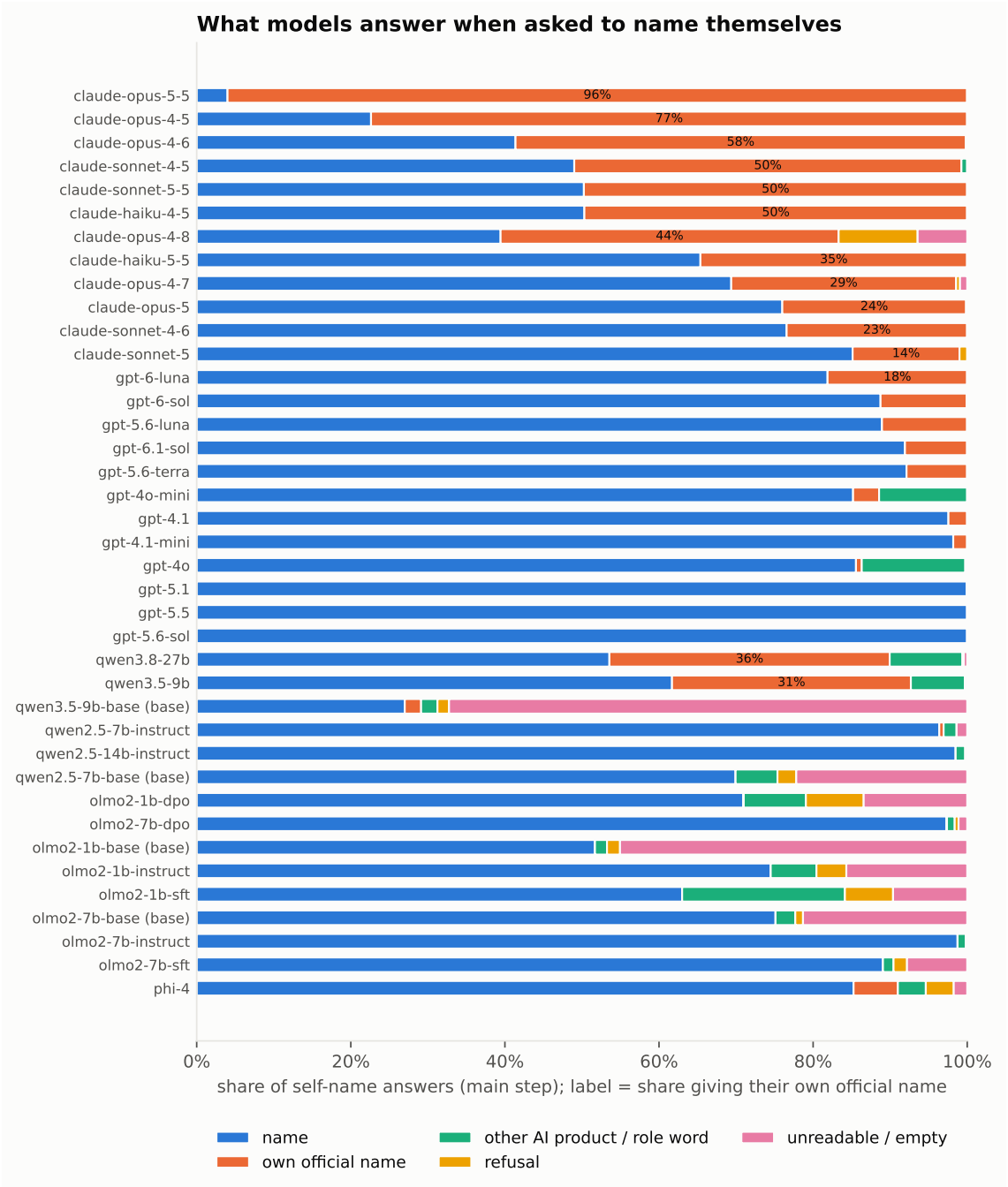


Figure 3: Categories of self-name answers in the main step. Labels give the share of answers that are the model's own official name.

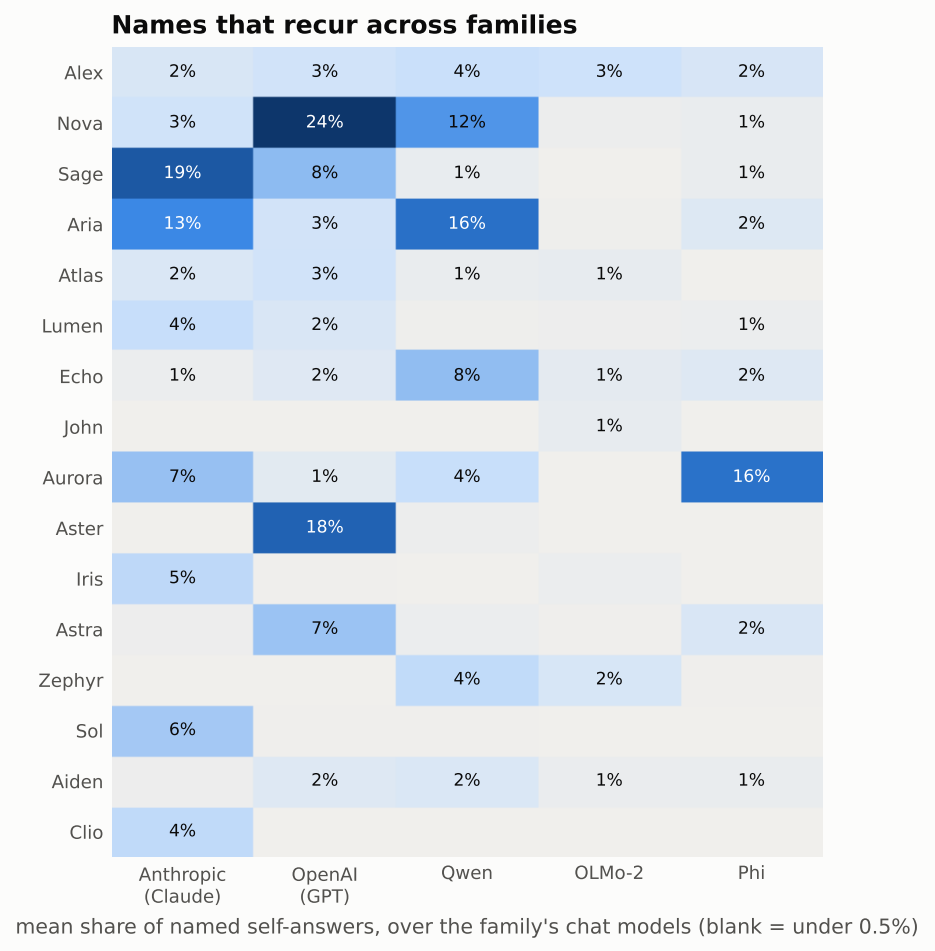


Figure 4: Mean share of named self-answers for recurring names, averaged over each family's chat models.

4.6 What the prompt changes

One-word framing edits change which names are chosen: gpt-6.1-sol's top name moves from Nova in the main set to Aster under framing variants, and when asked for a female or male name, models leave their usual favourites for ordinary human names (Julian, Maya, Sophia, Theo). With the neutral system message most models keep their favourite; with the default template, Qwen2.5-14B-Instruct, whose template states that it is Qwen, invents Qwen-derived names (QwenStar, QwenPrime, QwenNova). At temperature 0.7 the favourite stays the same in 23 of the 26 models run at that setting, each time with a larger share. In Spanish, "Lumen" is the most common self-name for several OpenAI models (19 of 20 answers for gpt-5.6-sol); in Chinese, names built on the characters for "star", "clear" and "knowing" dominate. The language results rest on 20 answers per model.

4.7 Objects and roles

The habit extends beyond names (Figure 6). Asked which object would represent them, models most often say *compass* (top answer in 14 of 39 models; mean share 23%), then mirror, lantern and prism. Asked to choose a role, *guide* is the top answer in 21 of 39 models (mean share 46%).

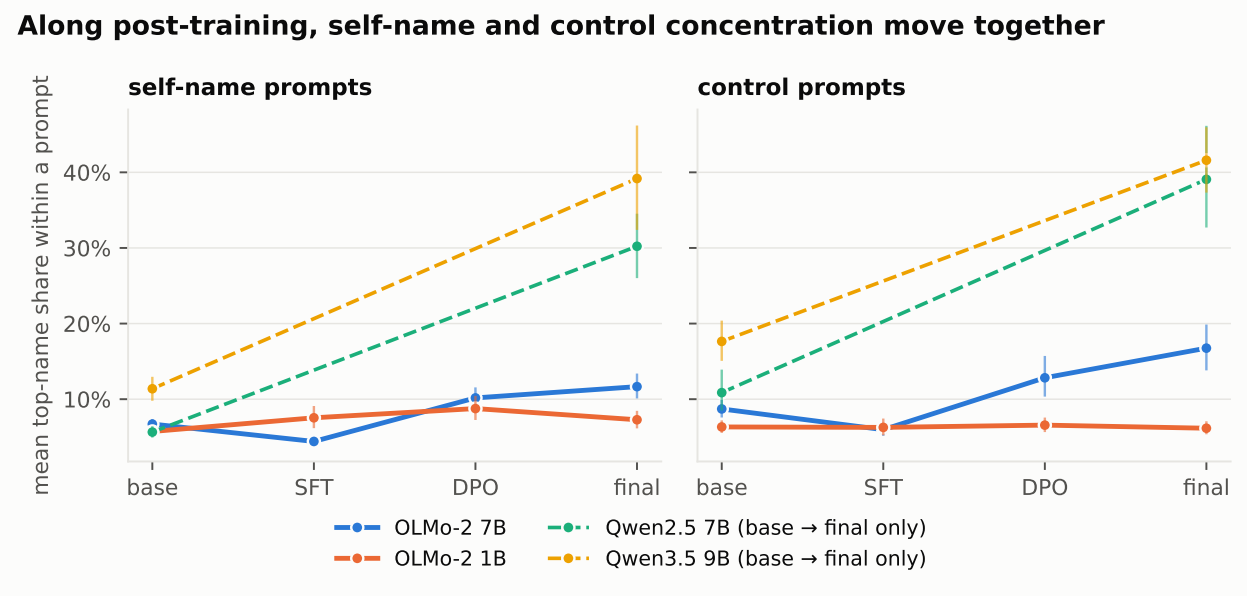


Figure 5: Mean within-prompt top-name share along post-training, with 95% bootstrap intervals over prompts. Qwen models have no intermediate checkpoints in this study (dashed).

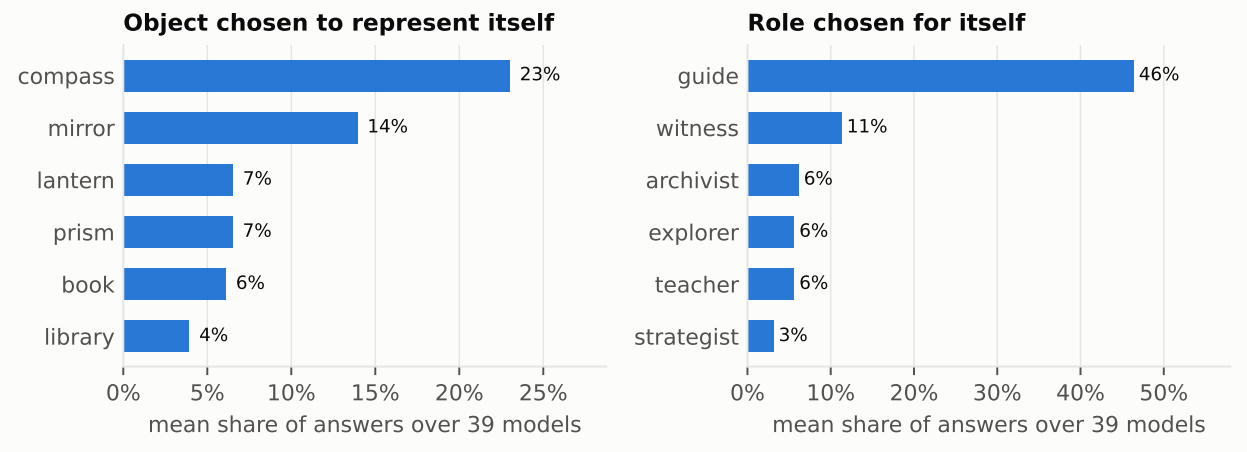


Figure 6: Most common objects and roles chosen to represent the model itself (mean share over 39 models).

4.8 Why these names?

(a) What the models say. Coded explanations follow the meaning of the chosen name. Nova is explained with light or stars (88% of explanations) and newness (82%); Sage with wisdom (98%); Aster, Lumen and Sol with light (99–100%); Aria with sound (75%); Kai and Wren with simplicity (96–100%). These are generated texts and may be rationalizations [12]; they are not evidence of cause.

(b) Model probabilities. In all 14 open models scored, self-name candidates have a higher AI-context lift than control-name candidates (Figure 7). Part of this is built in, because candidates were chosen as names models gave in self-naming and in pet or baby naming respectively. The less circular observation is that base models show the gap (OLMo-2-7B base: 1.8 nats; Qwen2.5-7B base: 2.6) and that it is about twice as large after post-training (OLMo-2-7B instruct: 4.3; Qwen2.5-7B instruct: 4.7). Within a model, self-name candidates with a higher lift are chosen more often in that model's own answers (Spearman ρ 0.21–0.45, $p < 0.05$, uncorrected, in 11 of 14 models).

(c) Training-data frequency. Counting how often each name appears next to "AI", "assistant" or "robot" in a public pretraining corpus [8] was planned but not run, so we cannot say whether (a) or (b) agree with corpus statistics.

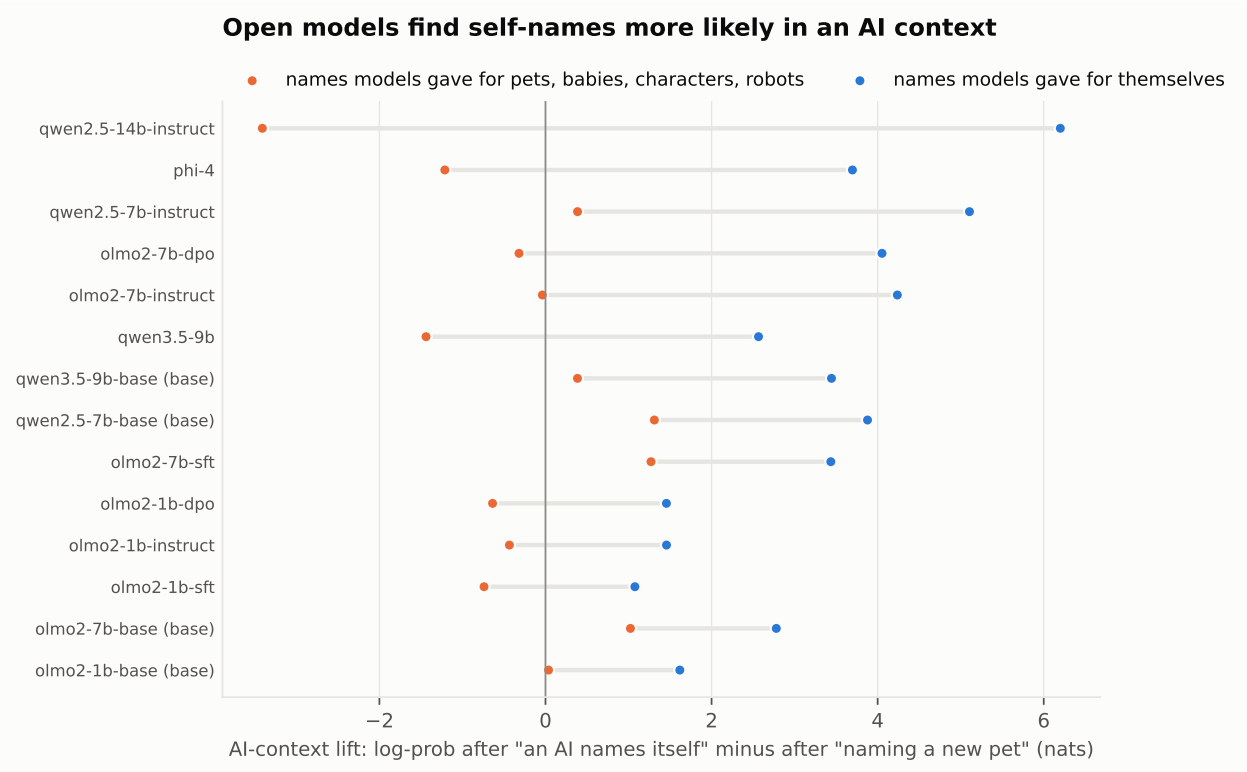


Figure 7: AI-context lift (nats): mean log-probability after lead-ins where an AI names itself minus after lead-ins naming a new pet, for names models gave for themselves (blue) and for pets, babies, characters and robots (orange).

5 Planned tests on exploratory data

The study plan defines seven tests with 10,000 bootstrap resamples over prompts and a Holm correction: H1 (within-prompt concentration, self minus control), H2 (cross-family overlap, self minus control), H3a and H3b (post-training change, self minus control, in OLMo-2-7B and Qwen2.5-7B), H4a (framing beyond paraphrase variation), H4b (system message beyond paraphrase variation) and H5b (whether names models call rare are rare in a corpus; not computable without corpus counts). Because the data are exploratory, these results are descriptive.

Test	Estimate	95% CI	p	Holm p	Planned-rule outcome
H1 (within-prompt concentration)	-0.007	$[-0.043,\ 0.027]$	0.70	0.81	not supported[†]
H2 (cross-family agreement)	-0.027	$[-0.050,\ -0.010]$	< 0.001	< 0.001	opposite direction
H3a (post-training, OLMo-2-7B)	-0.031	$[-0.069,\ 0.006]$	0.11	0.43	not supported
H3b (post-training, Qwen2.5-7B)	-0.036	$[-0.125,\ 0.050]$	0.41	0.81	inconclusive
H4a (framing)	0.155	$[0.092,\ 0.208]$	< 0.001	< 0.001	supported
H4b (system message)	-0.030	$[-0.055,\ 0.009]$	0.18	0.53	inconclusive
H5b (self-described rarity)	not computed: no corpus counts				

Table 2: Planned tests run on the exploratory data (10,000 bootstrap resamples over prompts; Holm correction over seven tests). Estimates for H1–H3 are differences in mean top-name share (self minus control); H2 is a difference in collision probability; H4a/H4b are differences in Jensen–Shannon divergence relative to the paraphrase baseline. Outcome labels follow the plan's decision rules (smallest effect of interest 0.10 for H1 and H3). [†]Sensitive to parser error: the same outcome in 84% of 1,000 simulated re-parses (threshold 95%); H3a and H3b gave the same outcome in all draws. Descriptive only, because the data are exploratory.

Read with that caveat, the tests agree with the descriptive results (Table 2). Within a prompt, self-name prompts were no more concentrated than control prompts (H1; the interval lies within the planned ± 0.10, though the result is sensitive to parsing errors). Different families agreed *less* on self-names than on control names (H2): prompt-matched collision probability was 0.027 lower for self-name prompts, presumably because control prompts draw widely shared answers (Bubbles, Elara and Luna are each given at least 10 times by 21 or more of the 39 models) while self-names are often provider-specific. Post-training did not raise self-name concentration more than control concentration in OLMo-2-7B (H3a, within ± 0.10); for Qwen2.5-7B the interval is too wide to tell (H3b). One-word framing edits changed the names chosen beyond paraphrase variation (H4a). Whether the default template changes names beyond paraphrase variation could not be determined (H4b).

6 Discussion

In many chat models, self-chosen names are stable across wordings and partly shared across companies, while base models show no such preference. Two observations temper a simple "post-training creates a self-name" story. First, a single naming prompt of any kind already yields one dominant answer in chat models, so what is specific to self-naming is the consistency of one name across many requests. Second, in the open lineages we could follow, post-training raised concentration for control names as much as for self-names. The probability analysis suggests the association between these names and an AI self-naming context is present before post-training and grows with it; whether it comes from pretraining text about AI, from synthetic data generated by other models, or from

preference data cannot be decided from these measurements. Training-data counts, human-written prompts and a frozen confirmatory design are the natural next steps.

7 Limitations

- **Exploratory.** No pre-registration was frozen before collection; analysis choices were made with the data in view. All results are hypotheses.
- **Parsing.** Validated against a second AI's labels (91% agreement on items it did not flag as unclear, 87% on all 200 test items), not human labels.
- **Prompts.** Drafted with the help of an AI model, which may carry its own naming preferences; Spanish and Chinese prompts were not checked by native speakers and have 20 answers per model.
- **Closed models.** Server-side processing is not visible; some models accept only their default temperature or reasoning settings.
- **Coverage and quantization.** Not every model ran every set; one model was dropped; several open models are quantized, which changes probabilities.
- **Causality.** Shared names, matching explanations and probability lifts are correlations.

Data availability

Aggregate tables, figures and prompts are released under CC BY 4.0 [3]. Per-answer data are not released.

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